분포 반전: 3준위계

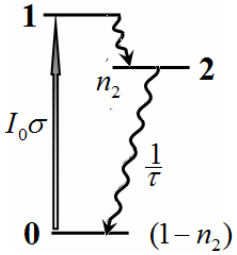
3.4 .

2 :

$$\frac{dn_2}{dt} = -\frac{n_2}{\tau} + I_0\sigma\left(1-n_2\right). \qquad (4)$$

:

2 0.



3.5 $dn_2/dt = 0$, (4)

$$\bar{n}_2 = \frac{I_0\sigma}{\frac{1}{\tau} + I_0\sigma} = \frac{I_0\sigma\tau}{1+I_0\sigma\tau}. \qquad (5)$$

$$\Delta n = \bar{n}_2 - \bar{n}_0 = \bar{n}_2 - \left(1-\bar{n}_2\right) = \frac{I_0\sigma\tau - 1}{1+I_0\sigma\tau}. \qquad (6)$$

3.6 , $\Delta n > 0$.

(6)

$$I_0\sigma\tau > 1. \qquad (7)$$

분포 반전: 4준위계

3.7 4 2

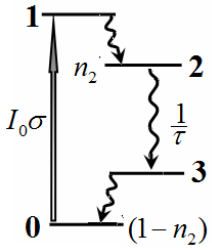
(4) ,

(5) .

2 3 ,

0 .

$$\Delta n = \bar{n}_2 = \frac{I_0\sigma\tau}{1+I_0\sigma\tau}, \qquad (8)$$



2 3 .

,

$$I_0\sigma\tau > 0. \qquad (9)$$

공진기

3.8 dN .

dt

$$dN_{out} = \left(1-\rho\right) I_G S dt = -dN. \qquad (10)$$

S .

I_G

$\frac{N}{Sl}$ $\frac{c}{r}$

$$I_G = \frac{1}{2}\frac{N}{Sl}\frac{c}{r}. \qquad (11)$$

1/2

.

$$N = \frac{2rSl}{c} I_G \qquad (12)$$